TEMS Academy of Olympiad Math
Adhoc FE Problem Set

Md. Fuad Al Alam

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Problem 1. Let $f(x) = x^{67-x^{67-x^{67-\cdots}}}$, where $x \neq 0$. If $f(n) = 64$, then what is the remainder if n^n is divisible by 11?

Problem 2. Let

$$f(x) + 2f(-x) = 2x$$

and

$$2f(x) + f(-x) = x^2$$

How many x are there between 1 and 2024 for which both x and $f(x)$ are integers?

Problem 3. Let

$$f(x+y) = f(x) + f(y)$$

If $f(1) = 3$, find the value of $f(30)$.

Problem 4. Let $f(x,y) = y - x^2$, where $|x| + |y| \leq 21$. Find the difference between the largest and the smallest values of the function. (x and y both are integers)

Problem 5. If $f(n) = f(n+1) + f(n-1)$, $f(1) = 5$, $f(2) = 4$, then $f(2024) = ?$

Problem 6. Let $f(1) = 2, f(2n) = f(2n-1) + 1$ and $f(2n-1) = f(2n-3) + 2$ (when n is natural number). Find $f(2023)$.

Problem 7. If $f(x+3) = f(x+1) + 4x + 12$ and $f(0) = 2$, then what is $f(500)$?

Problem 8. Let $S = \{1, 2, 3, \ldots, 12\}$. How many functions $f : S \to S$ are there such that $f(f(x)) = x$ and $f(x) - x$ is divisible by 3?

Problem 9. If $f(x) = 3x - 2$, then what is the value of x in the following equation where f is 2020 times iterated?

$$x = f(f(f(\ldots f(x) \ldots)))$$

Problem 10. If $f(n) = f(n-1) + 2n - 2$ and $f(1) = 2$, then what is the value of $f(100)$?

Problem 11. Let

$$f(3n) = 3f(n)$$
$$f(3n+1) = 3f(n) - 1$$
$$f(3n+2) = 3f(n) - 2$$

If $f(1) = f(2) = 1, f(1009) =$?

Problem 12. If $f(x) = 2019$, then what is the value of $f^{1019}(2019)$? **Note:** $f^2(x) = f(f(x))$.

Problem 13. Two functions $f(x, y)$ and $g(x)$ are defined on real numbers, such that

$$f(x+y, x-y) = x \cdot g(y) + y \cdot g(x)$$

Given that, $g(x) = 5$ for all values of x. Find the value of $f(2019, 0)$.

Problem 14. Let $f(1) = 2, f(2) = 5, f(3) = 10$ and

$$f(x) - f(x-3) = 2[f(x-1) - f(x-2)]$$

where $x > 3$. $f(2019) =$?

Problem 15. Let $f(4x-3) = 2x + 5$. If $f(2x+1) = ax + b$, $a + b =$?

Problem 16. Suppose $f(x)$ is a function. It is given that

$$f(2n-1) = f(2n-2) - 1$$

and

$$f(2n) = 2f(2n-1)$$

If $f(1) = n$, then what is the value of n such that $f(1971) - f(2018) = -1$?

Problem 17. If $f(x^2 - 2016x) = xf(x) + 2016$, what is the value of $f(2017)$?

Problem 18. Let

$$f(x) = g(x+1)$$
$$f(y) = 2018y + 2016y + \ldots + 2y$$
$$g(x) = h(2x) - x$$

If $h(2018) = a^3$ and a is an integer, $3a =$?

Problem 19. For any rational numbers x, y, function $f(x)$ is a real number and

$$f(x+y) = f(x)f(y) - f(xy) + 1$$

Again, $f(2017) \neq f(2018)$, $f(\frac{2017}{2018}) = \frac{a}{b}$, where a, b are co-prime. $a + b =$?

Problem 20. If $F_0(x) = \frac{1}{1-x}$ and $F_n(x) = F_0(F_{n-1}(x))$. Find $F_{2015}(2016)$.

Problem 21. Let

$$f(3m) = \frac{mf(m)}{3}$$
$$f(3m+2) = \frac{(m+2)f(m+2)}{3}$$
$$f(3m+1) = \frac{(m+1)f(m+1)}{3}$$

$f(2016) =$?

Problem 22. For all positive integers x, $f(f(x)) = 4x + 3$ and for one positive value of integer k, $f(5^k) = 5^k \times 2^{k-2} + 2^{k-3}$. $f(2015) =$?

Problem 23. Let $f(y) = y$ repeated y times, for example $f(3) = 333, f(5) = 55555$. If $a = f(2001) + f(2002) + \ldots + f(2014) + f(2015)$. What is the remainder upon division of a by 3?

Problem 24. $f(n) =$ sum of the squares of digits of n. If $f_k(n) = f(f_{k-1}(n))$, $f_{2014}(3) =$?

Problem 25. For all positive integers x, y, $f(x) \geq 0$ and $f(xy) = f(x) + f(y)$. If the digit at the one's of x is 6, then $f(x) = 0$. If $f(1920) = 420$, then $f(2015) =$?

Problem 26. If x, a, b are positive integers, $f(x)$ is positive integer too, and if $a > b$, then $f(a) > f(b)$. Again, $f(f(x)) = x^2 + 2$. Find the value of $f(6)$.

Problem 27. A function $f : \mathbb{R} \to \mathbb{R}$ is defined in such a way that $f(x+y) = f(x) \cdot f(y)$, and for $a \in \mathbb{N}$, $\sum_{k=1}^{n} f(a+k) = 16(2^n - 1)$. If $f(1) = 2$, then what is the value of a?

Problem 28. Given that, $F(x, y) = F(xy, \frac{x}{y})$ where $y \neq 0$. If $F(x^4, y^4) + F(x^2, y^2) = 16$, then $F(x, y) =$?

Problem 29. If $f(x, y) = x + y \times f(y, x)$, then $f(2, 1) =$?

Problem 30. A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x+7) \geq f(x) + 7$ and $f(x+1) \leq f(x) + 1$. $f(2013) =$?

Problem 31. Let $f : \mathbb{R} - \{0\} \to \mathbb{R}$ be a function such that $f(xy) = \frac{f(x)}{y}$. If $f(2012) = 1$, then $f(2013) =$?

Problem 32. The function $f(x)$ satisfies the equation

$$(x-2)f(x) + f(\frac{1}{x}) = 1$$

Find the value of $f(3)$.

Problem 33. If $f(x) = a + bx$ and $f(f(f(x))) = 27x + 52$, and if a and b are real numbers, then $a - b =$?

Problem 34. What is the domain of the function $f(x) = \log_{|x|} \sqrt{x^2}$?

Problem 35. For an injective function $f : \mathbb{R} \to \mathbb{R}$,

$$f(x + f(y)) = 2012 + f(x + y)$$

$f(2013) =$?